

Hospital Admissions Dataset

Working Spec & Practice Questions

1. Purpose

This spec covers what you need to work with the hospital admissions practice dataset, and the questions you should be able to answer from it by the end of the exercise. The data is entirely synthetic, no real patients and contains intentional duplicate admission records so you can build and test deduplication logic.

2. Dataset Overview

File: hospital_admissions_raw.csv 108 rows: 100 unique patients plus 8 duplicate admission entries (the same admission captured twice, e.g. once at reception and again at the ward).

Column	Description
patient_id	Synthetic patient identifier (e.g. PAT0001)
first_name / surname	Fabricated name
national_id	Fabricated ID number, SA ID shape only — not a real or valid number
date_of_birth	Fabricated date of birth
disease	Diagnosis at admission
severity	low / medium / high
admission_status	Admitted / Not Admitted
ward	Ward assigned if admitted, else N/A
admission_date	Date the admission event occurred
discharge_date	Date discharged, blank if not admitted

3. Prerequisite Skills

- Core SQL: SELECT, JOIN, GROUP BY, window functions (ROW_NUMBER, PARTITION BY).
- Understanding of natural/business keys vs. surrogate keys — this dataset's natural key is (patient_id, admission_date), not patient_id alone.
- Comfort reading a MERGE / UPSERT (ON CONFLICT DO UPDATE) statement.
- Basic date arithmetic (age from date_of_birth, length of stay from admission/discharge dates).

4. Tooling & Access Checklist

Requirement	Why it's needed
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A SQL engine (Postgres, Snowflake, or SQL Server)	Runs the schema, dedup/merge script, and your own queries
Ability to load a local CSV into a table	Loads hospital_admissions_raw.csv as is, duplicates included
A query tool / IDE	DBeaver, VS Code + SQL extension, or your platform's native console

5. What to Do With the Data

1. Load hospital_admissions_raw.csv into hospital_admissions_raw exactly as provided. Do not pre-clean it.
2. Run the dedup/merge script from hospital_pipeline.sql into hospital_admissions.
3. Confirm the curated table has exactly 100 rows and no duplicate (patient_id, admission_date) pairs.
4. Answer the practice questions in Section 6 using the curated table.

6. Practice Questions

A. Data quality & deduplication

5. How many rows landed in the raw table, and how many remained after dedup? What does the difference tell you?
6. Write a query that finds every duplicate admission entry in the raw table before you dedupe it.
7. Why is (patient_id, admission_date) the correct natural key here instead of patient_id alone? Give a scenario where deduping on patient_id alone would silently delete a legitimate admission.
8. One duplicate pair in this dataset differs only by ward casing ("ICU" vs "icu"). What other kinds of near-duplicates (typos, whitespace, date format drift) could break a naive dedup query, and how would you guard against each?
9. Write a query to prove, after your merge runs, that no (patient_id, admission_date) pair appears twice.

B. Clinical / operational questions

10. What is the admission rate (Admitted vs Not Admitted) for each severity level (low / medium / high)?
11. Which disease has the highest number of admitted patients?
12. Which ward has the most admissions, and does that match what you'd expect given the severity mix?
13. What is the average length of stay (discharge_date – admission_date) by severity level?
14. Are there any patients marked 'Not Admitted' despite having a high severity disease? Is that a data quality issue, a business rule exception, or something to flag to a clinician?
15. What is the age (from date_of_birth) distribution of admitted patients? Is any age group over or underrepresented?

C. Pipeline design

16. How would you automate this pipeline to run every time a new CSV of admissions lands? What would trigger a run?
17. How would you detect a new kind of duplicate you haven't seen yet (e.g. same national_id but a different patient_id — a possible registration error) before it reaches the curated table?
18. If this pipeline were pointed at real patient data tomorrow, what would need to change or be reviewed first?

7. Governance Note

This dataset is synthetic and for training only. If this pipeline pattern is ever applied to real patient data, it must go through proper data governance and privacy review before use, and any official reporting figures should come from your organization's governed reporting source rather than an ad hoc pipeline like this one.